



CondorBrain™-CondorIntelligence™ Architecture Diagrams – Reference Guide

This document describes the purpose and scope of each architecture diagram used in the CondorBrain™-CondorIntelligence™ system. These diagrams collectively define the structural, computational, and decision-making design of an institutional-grade SPY options trading platform for benchmarking retail automated trading strategies.

1. Master Architecture Maps

diagram_map.png

The master navigational diagram for the entire CondorBrain system. This map organizes all other diagrams into functional clusters, including Data, Intelligence, Logic, Execution, and Optimization. It serves as the index and entry point for understanding system structure.

system_overview.png

A high-level conceptual overview illustrating how the Data layer feeds the Intelligence Core, which in turn drives the Execution layer. This diagram is intended for executive, audit, and stakeholder understanding rather than implementation detail.

full_system_architecture.png

The authoritative V2.2 system specification diagram. This diagram provides a complete, end-to-end view of the CondorBrain architecture, including the Mamba-2 neural backbone, Diffusion forecasting components, decision logic, and execution pathways. It is the primary technical reference for the system.

phase2_phase3_architecture.png

A roadmap diagram outlining the architectural evolution from Phase 2 (indicator-driven logic

and baseline neural intelligence) to Phase 3 (advanced risk controls, regime awareness, institutional execution, and adaptive optimization).

2. Intelligence Core Diagrams

enhanced_architecture.png

A focused view of the Neural Intelligence Core, highlighting the Mamba-2 backbone and its connections to specialized output heads, including Diffusion-based forecasting and Top-K Mixture-of-Experts (MoE) routing.

mamba_engine_logic.png

An internal schematic of a single Mamba-2 block. This diagram illustrates the Selective Scan State Space Model (SSM), gating mechanisms, and hardware-accelerated computation paths used to efficiently process long financial sequences.

condor_intelligence_flow_premium.png

An enhanced intelligence flow diagram that extends the standard neural pipeline with Topological Data Analysis (TDA) and Generative Diffusion components. This diagram reflects the premium research configuration used for advanced regime detection and probabilistic forecasting.

condor_intelligence_flow.png

The standard intelligence flow diagram showing how normalized input tensors are transformed through neural layers into actionable trading signals and forecasts.

3. Data Pipeline Diagrams

pipeline_diagram.png

A visual representation of the Extract-Transform-Load (ETL) pipeline responsible for ingesting raw market and options data from external providers (e.g., Alpaca, IVolatility) and preparing it for downstream processing.

data_pipeline_detailed.png

A detailed view of data ingestion and transformation, focusing on the construction of the MarketSnapshot object and precise feature alignment across price, volatility, and options-specific inputs.

lag_alignment_flow.png

A critical synchronization diagram illustrating how delayed or lagged data sources (e.g., 15-minute indicators or options greeks) are correctly aligned with real-time price data to prevent leakage and temporal inconsistencies.

4. Decision Logic and Position Sizing

dataflow.png

A comprehensive mapping of all technical indicators, hard gates, and rule-based filters, showing how they feed into the fuzzy logic inference system.

fuzzy_sizing_pipeline.png

A detailed diagram of the fuzzy logic engine that computes the Fuzzy Confidence Score. This score directly influences position sizing and risk exposure.

membership_curves.png

Visual representations of fuzzy membership functions (e.g., RSI, volatility, VIX proxies) that normalize raw indicator values into bounded confidence scores between 0 and 1.

position_sizing.png

Illustrates the risk management and sizing logic, including Kelly Criterion-inspired calculations and capital allocation constraints that determine the number of option contracts to trade.

entryexitdecision.png

A logic tree diagram describing the decision-making process for entering and exiting Iron Condor positions based on signal strength, confidence thresholds, and market conditions.

exit_priority.png

Defines the hierarchy of exit conditions, ranking profit targets, stop losses, volatility violations, and technical invalidations (such as Bollinger Band breakouts) by priority.

5. Execution and Optimization

institutional_execution_flow.png

Describes the end-to-end execution workflow, including broker interaction, order lifecycle management, pre-trade risk checks, and post-trade validation.

optimization_pipeline.png

Illustrates the feedback loop connecting model performance, backtesting results, and live execution metrics back into retraining, hyperparameter tuning, and system optimization.

CondorBrain™-CondorIntelligence™

Architecture – Figure Cross-Reference Table

Figure No.	File Name	Figure Title / Description
Figure 1	diagram_map.png	Master Navigational Map of CondorBrain Architecture
Figure 2	system_overview.png	High-Level System Overview: Data, Intelligence, and Execution Layers
Figure 3	full_system_architecture.png	Full CondorBrain System Architecture (V2.2 Reference Specification)
Figure 4	phase2_phase3_architecture.png	Phase 2 → Phase 3 Architectural Roadmap
Figure 5	enhanced_architecture.png	Enhanced Intelligence Architecture with Mamba-2, Diffusion, and Top-K MoE
Figure 6	mamba_engine_logic.png	Internal Mamba-2 Engine Logic and Selective Scan SSM
Figure 7	condor_intelligence_flow.png	Standard Neural Intelligence Flow
Figure 8	condor_intelligence_flow_premium.png	Premium Intelligence Flow with TDA and Generative Diffusion
Figure 9	pipeline_diagram.png	Market Data ETL and Processing Pipeline
Figure 10	data_pipeline_detailed.png	Detailed Data Ingestion and MarketSnapshot Construction
Figure 11	lag_alignment_flow.png	Lag and Timestamp Alignment Logic for Delayed Data
Figure 12	dataflow.png	Indicator and Signal Dataflow into Decision Logic
Figure 13	fuzzy_sizing_pipeline.png	Fuzzy Logic Inference and Confidence Scoring Pipeline
Figure 14	membership_curves.png	Fuzzy Membership Functions and Normalization Curves
Figure 15	position_sizing.png	Position Sizing and Risk Allocation Logic
Figure 16	entryexitdecision.png	Trade Entry and Exit Decision Logic
Figure 17	exit_priority.png	Exit Condition Priority Hierarchy
Figure 18	institutional_execution_flow.png	Institutional-Grade Trade Execution Workflow
Figure 19	optimization_pipeline.png	Training, Feedback, and Optimization Loop